Database Management Systems

Lecture 01- Introduction

Data and Information

- ✓ Data is a collection of **raw facts, figures, or observations** that may not have a meaningful interpretation by itself.
- ✓ Examples:
  - 75, 80, 90, 65
  - Rahim, Karim, Sakib
  - 25°C
  - 10101001

- ✓ Information is **processed and organized data that has meaning and can be used for decision-making**.

- ✓ **Example in real life:**
  If a student's marks are **70, 85, 90, and 75**, calculating them gives an **average mark of 80**. This average is information.

Database & DBMS (Database Management System)

A **database** is an organized collection of **related data** that can be easily stored, accessed, managed, and updated.

Example:

A university database may store student information:

Student ID	Name	Department	CGPA
101	Rahim	CSE	3.50
102	Karim	CSE	3.75
103	Sakib	EEE	3.40

- ✓ **DBMS (Database Management System)** is software that allows users to **create, store, manage, retrieve, update, and delete data** in a database.
- ✓ **Examples of DBMS**
  - MySQL
  - Oracle Database
  - Microsoft SQL Server
  - PostgreSQL
  - SQLite

Data Models, Schema and Instances

A **data model** is a set of concepts used to describe the **structure, relationships, and constraints of data** in a database.

Suppose we have a university database:

STUDENT

Student_ID
Name
Department

COURSE

Course_ID
Course_Name

We can define a relationship between **Student** and **Course**:

STUDENT ——— Enrolls ——— COURSE

This structure represents a **data model**.

Common types of data models

1. **Relational Model** – data is organized into tables.
2. **Hierarchical Model** – data is organized like a tree.
3. **Network Model** – data is represented using records and relationships.
4. **Object-Oriented Model** – data is represented as objects.

Schema

- ✓ A **schema** is the **overall logical structure or design of a database**.

- ✓ **Example**

  Suppose our Student table is designed as:

STUDENT

Student_ID	INT
Name	VARCHAR
Department	VARCHAR
CGPA	DECIMAL

- ✓ This structure is the **schema** of the Student table
- ✓ The schema generally **does not change frequently**.

Instance

- ✓ A **database instance** is the **actual data stored in the database at a particular point in time**.
- ✓ Schema:

  Student(Student_ID, Name, Department, CGPA)
- ✓ Current data:

Student_ID	Name	Department	CGPA
101	Rahim	CSE	3.50
102	Karim	CSE	3.75
103	Sakib	EEE	3.40

These actual records represent the **instance** of the database.

If a new student is added:

Student_ID	Name	Department	CGPA
101	Rahim	CSE	3.50
102	Karim	CSE	3.75
103	Sakib	EEE	3.40
104	Nadia	CSE	3.90

The **schema remains the same**, but the **instance changes**.

Difference at a Glance

Concept	Meaning	Example
Data Model	How data is structured and related	Relational model
Schema	Design/structure of the database	Student(ID, Name, Dept, CGPA)
Instance	Actual data at a particular time	101, Rahim, CSE, 3.50

Thank You!